

Hands-on: Debugging, Evaluating and Monitoring LLM Apps with LangSmith

Instructor

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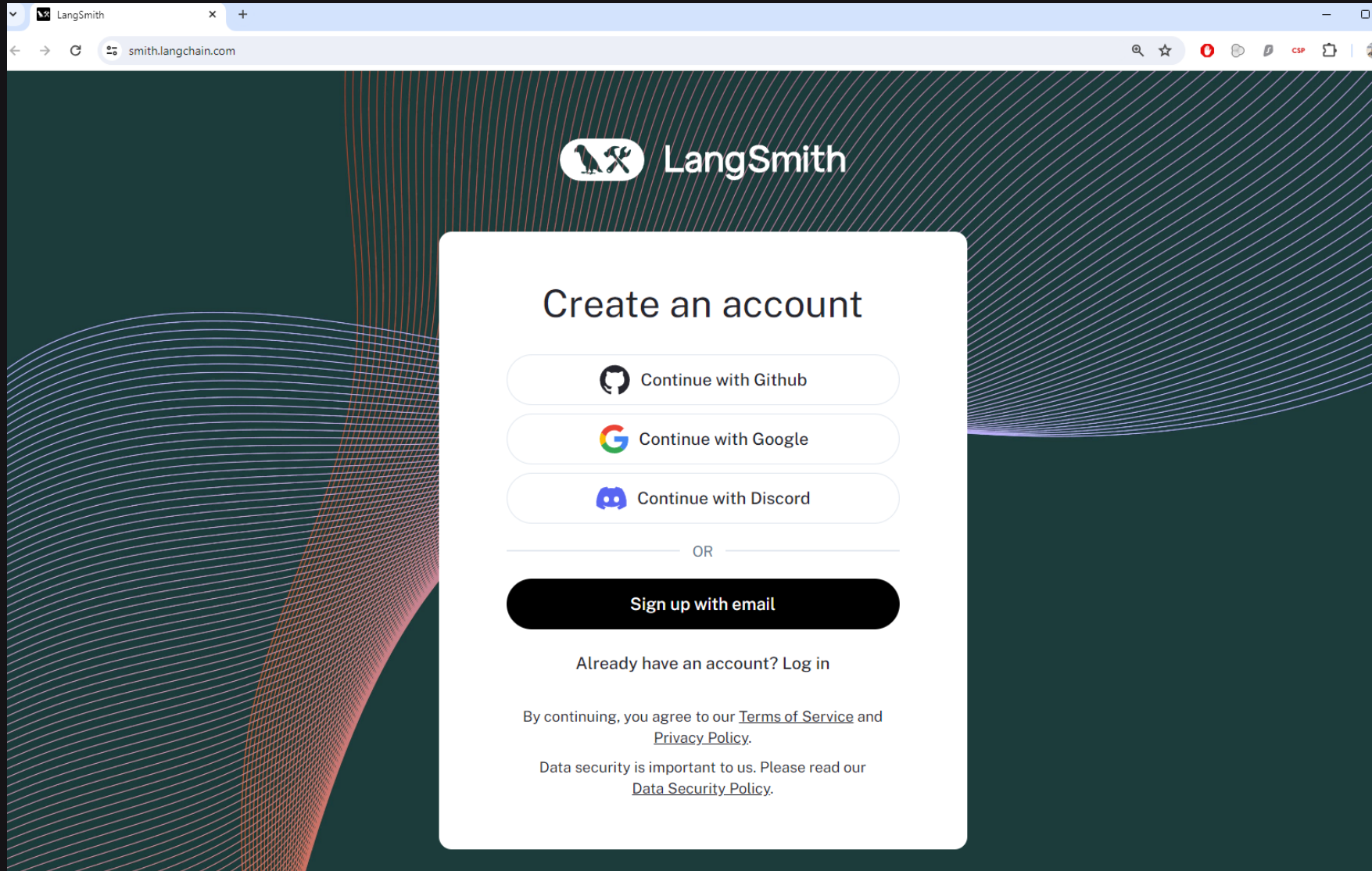
Head of Community & Principal AI Scientist at Analytics Vidhya

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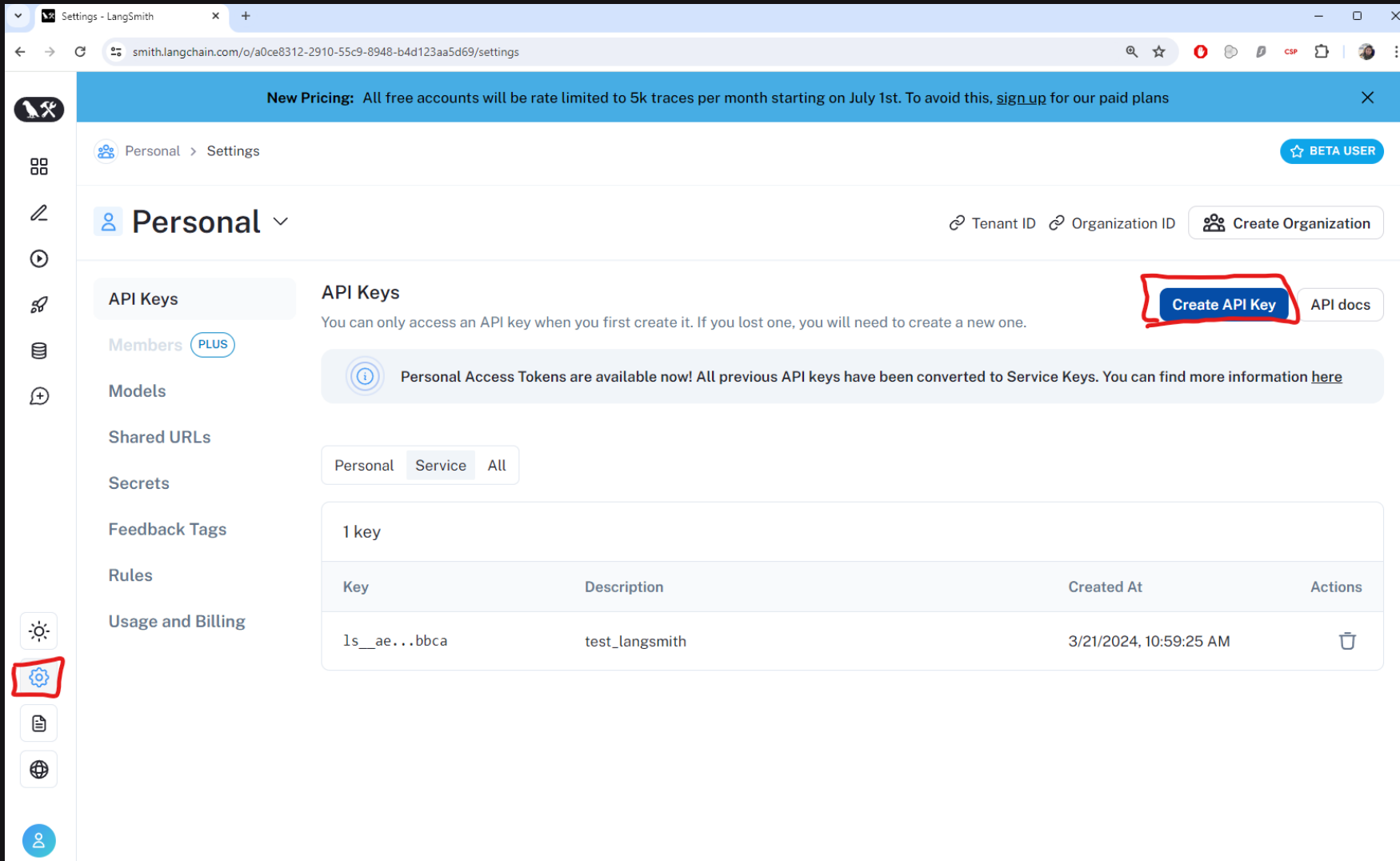
Published Author



Register an account on LangSmith

A screenshot of a web browser showing the LangSmith registration page. The browser's address bar displays 'smith.langchain.com'. The page features a dark green background with a pattern of thin, curved, light-colored lines. At the top center is the LangSmith logo, which consists of a circular icon with a cat and a wrench, followed by the text 'LangSmith'. Below the logo is a white rectangular card with rounded corners. Inside the card, the heading 'Create an account' is centered. Underneath the heading are three buttons for social login: 'Continue with Github' (with the Github logo), 'Continue with Google' (with the Google logo), and 'Continue with Discord' (with the Discord logo). Below these buttons is a horizontal line with the word 'OR' in the center. Underneath the line is a black button with white text that says 'Sign up with email'. Below the black button is the text 'Already have an account? Log in'. At the bottom of the card, there are two lines of text: 'By continuing, you agree to our [Terms of Service](#) and [Privacy Policy](#).' and 'Data security is important to us. Please read our [Data Security Policy](#).'

Create an API Key



Settings - LangSmith

smith.langchain.com/o/a0ce8312-2910-55c9-8948-b4d123aa5d69/settings

New Pricing: All free accounts will be rate limited to 5k traces per month starting on July 1st. To avoid this, [sign up](#) for our paid plans

Personal > Settings

BETA USER

Personal

Tenant ID Organization ID Create Organization

API Keys

Members PLUS

Models

Shared URLs

Secrets

Feedback Tags

Rules

Usage and Billing

API Keys

You can only access an API key when you first create it. If you lost one, you will need to create a new one.

Create API Key API docs

Personal Access Tokens are available now! All previous API keys have been converted to Service Keys. You can find more information [here](#)

Personal Service All

1 key

Key	Description	Created At	Actions
ls__ae...bbca	test_langsmith	3/21/2024, 10:59:25 AM	

Create an API Key

Personal > Settings

BETA USER

Personal

API Keys

You can only access API Keys that you own.

1 key

Key

1s__ae...

Created At

3/21/2024, 10:59:25 AM

Actions

ⓘ NOTE

The API key will be shown only once, so make sure to copy it and store it in a safe place.

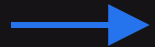
Start monitoring LLM App runs with LangSmith

```
3 os.environ['OPENAI_API_KEY'] = OPENAI_KEY
4 os.environ["LANGCHAIN_TRACING_V2"] = "true"
5 # you can change this based on specific apps and projects
6 os.environ["LANGCHAIN_PROJECT"] = f"Default Project"
7 os.environ["LANGCHAIN_ENDPOINT"] = "https://api.smith.langchain.com"
8 os.environ["LANGCHAIN_API_KEY"] = LANGSMITH_KEY # Update to your API key
```

```
[5] 16
    17 # create simple llm chain
    18 llm_chain = (prompt
    19             |
    20             chatgpt
    21             |
    22             StrOutputParser())
    23 )
```

```
▶ 1 prompt_txt = "Explain AI in 3 bullet points"
   2 response = llm_chain.invoke({'human_input': prompt_txt})
   3 print(response)
```

→ - AI, or artificial intelligence, refers to the simulation of
- It involves the development of algorithms and models that en
- AI technologies include machine learning, natural language p



The screenshot shows the LangSmith web interface for a project named "Default Project". The interface includes a sidebar with navigation options like "Runs", "Threads", "Monitor", and "Setup". The main area displays a trace for a "RunnableSequence" with a duration of 1.83s and 116 tokens. The trace shows the following components and their durations:

- ChatPromptTemplate: 0.00s
- ChatOpenAI (gpt-3.5-turbo): 1.81s
- StrOutputParser: 0.00s

The input for the trace is "human_input: Explain AI in 3 bullet points". The output is a list of three bullet points:

- AI, or artificial intelligence, refers to the simulation of human intelligence processes by machines, typically computer systems.
- It involves the development of algorithms and models that enable machines to perform tasks that typically
- AI technologies include machine learning, natural language p

Create LLM App Evaluation Datasets with Ease

```
1 dataset = client.create_dataset(  
2     dataset_name='Sample LLM Dataset - DJTest001',  
3     description="Dataset of sample prompts and outputs",  
4 )  
5  
6 for input_prompt, output_answer in refined_dataset:  
7     client.create_example(  
8         inputs={"question": input_prompt},  
9         outputs={"answer": output_answer},  
10        metadata={"source": "Wikipedia"},  
11        dataset_id=dataset.id,  
12    )
```



Sample LLM Dataset - DJTest001

Dataset of sample prompts and outputs

Dataset ID: ... Add Evaluator Add Example New Experiment

Experiments Pairwise Experiments Examples

Filters Base Split All +

Input	Output	Created At	Modified At
What's the main characteristic of amphibia...	They live both in water and on land	6/5/2024, 12:5...	6/5/2024, 12:5...
What do mammals and birds have in comm...	null	6/5/2024, 12:5...	6/5/2024, 12:5...
What is the largest mammal?	The blue whale	6/5/2024, 12:5...	6/5/2024, 12:5...
Explain AI in one line	AI, or artificial intelligence, refers to the simulation of human intelligence processes by ...	6/5/2024, 12:5...	6/5/2024, 12:5...
Explain Generative AI in one line	Generative AI is a type of artificial intelligence that can create new content, such as ima...	6/5/2024, 12:5...	6/5/2024, 12:5...

Evaluate LLM Apps easily with evaluators

```
1 correct_evaluator = LangChainStringEvaluator("labeled_criteria",
2                                           config={"criteria": "correctness"})
3 conciseness_evaluator = LangChainStringEvaluator("criteria",
4                                                  config={"criteria": "conciseness"})
5 helpfulness_evaluator = LangChainStringEvaluator("criteria",
6                                                  config={"criteria": "helpfulness"})
7 semantic_evaluator = LangChainStringEvaluator("embedding_distance")
8
9 client = Client()
10
11 results = evaluate(
12     lambda x: llm_chain.invoke({'human_input' : x['question']}),
13     client=client,
14     data="Sample LLM Dataset - DJTest001",
15     experiment_prefix="test_eval001",
16     evaluators=[correct_evaluator, conciseness_evaluator,
17                semantic_evaluator, helpfulness_evaluator]
18 )
```

